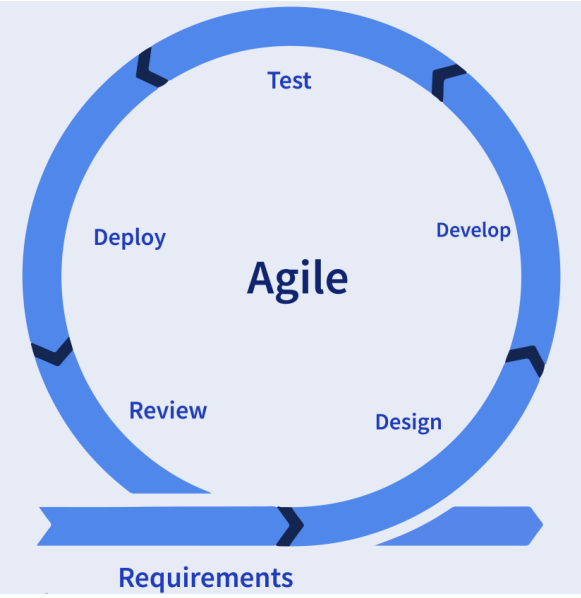
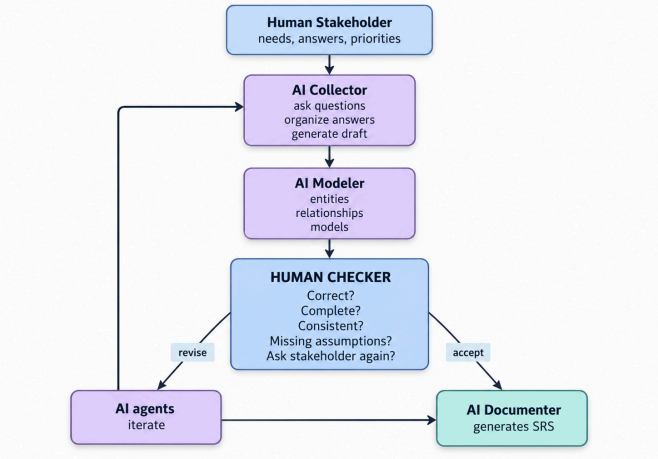
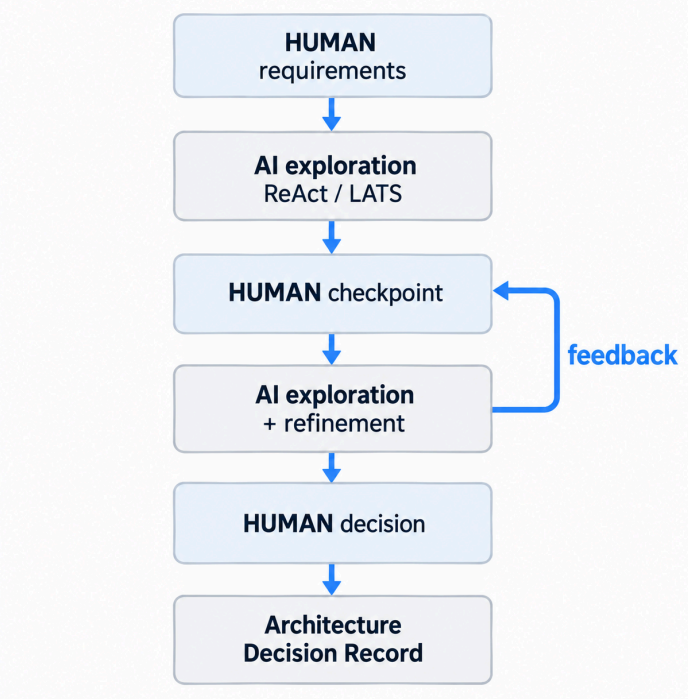
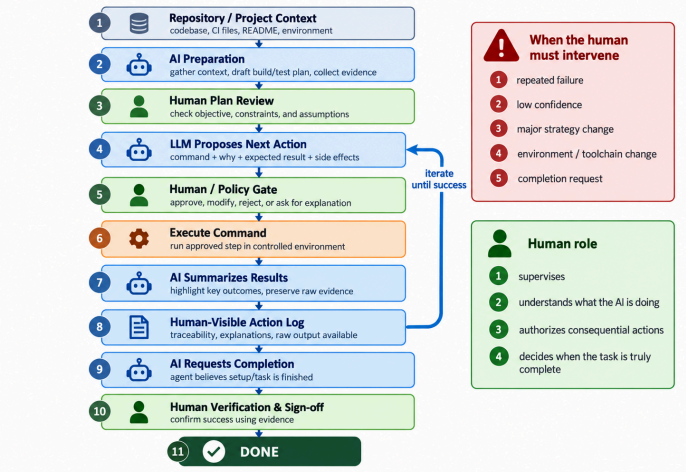
Our process model will follow an agile base. This gives the process model ample flexibility to respond to changing requirements.



What we need to do is to design each step so that the agentic AI isn't allowed to just go rogue, and complete every step without human knowledge. While the agent should be able to perform various tasks automatically, the output needs to be reviewed and tweaked by human operators before moving to the next step. Thus, we will design a specific process that must be followed at each stage.

Step	Source	Description
Requirements	https://arxiv.org/abs/2405.03256	The MARE paper describes a process by which agentic AIs collaborate. We can modify the loop by having the human take the place of the AI checker. Which forms our requirements loop:  This maintains the speed advantage of the AI.

		However, since the human has to be cleared on the critical details, we ensure that the human is still being briefed on what is happening.
Design	https://doi.org/10.1109/ICSA-C65153.2025.00015	This paper lays out an AI-based process which essentially goes: Find patterns Compare Patterns Make Decision Asses Decision Reflect (Judge)  All those stages are handled by agentic AI agents, and only the output choice is given to the human. I think the human should be much more pivotal to this process. Not just a “yes man” to the AI.
Develop	This step seemingly has poor research regarding process models.	I assume the reason why there aren’t any good papers modeling this stage is that the LLMs are so good at it from the start. For this, we need to derive our own way of thinking. For now, we will assemble a “double double” implementation strategy to minimize mistakes and automate the process. We instance two agents on the implementation of the same task. Once the two agents have completed the task, we dispatch two

		more separate agents to review each change independently. They are allowed to give either option a score between 1 and 3. 1 being bad, 2 being okay, 3 being good. These recommendations are passed along, to the human reviewer, who makes any changes, before accepting one of the two solutions. The review process is intended to quickly catch bugs and bad designs. Saving the human the effort of doing a more in-depth analysis.
Test	https://doi.org/10.1145/3728922	This paper is inherently what we are trying to address. It tries to perform full automation here. Keeping the humans entirely out. We want to introduce humans back into this loop, while keeping the best parts of what makes this so efficient. It would be foolhardy to replace the summarizer or control center, as those are incredibly tedious, and would benefit from automation. Better to add a human supervisor and verification between the LLM and execution. 
Deploy		Deployment should be handled manually. This is one thing that has been shown to be disastrous in the past, if done improperly. Additionally, due to the nature of deployment, there are significant security concerns. Humans should do this. AI should be available to assist only for technical questions on the deployment of the software.
Review		Review involves stakeholders and demos. New

		requirements are facilitated after review using the same process as before.

Update on 09/30/2026:

For the development phase:
There is an important weakness in the current “double-double” design. This process generates two implementations by 2 agents, then hands both to other agents for evaluation, and only afterward involves the human. **So we are handing over 2 agents' work to another 2 agents!!**

Drawbacks of AI reviewing AI

Reducing review to **1–3 scores loses information**. A human sees “Agent A = 3, Agent B = 2,” but not necessarily *why* A is safer, which tests failed, what assumptions differ, or what architectural compromises were made. This can create **automation bias**: the human may naturally focus on the AI-preferred solution rather than independently examining the important differences.

Most importantly, it puts the human **too late in the process**. By the time the human participates, four agents have already spent effort following potentially incorrect assumptions. This conflicts somewhat with our project's goal that humans should know what is happening and review/tweak AI output before proceeding.

I think we can break down the specific tasks and put one human checkpoint after critical decisions and **before expensive implementation begins**. With this approach, **we can also save computational cost.**

The steps can be like below:

Task
↓
Agent A + Agent B independently propose implementation plans
↓
Human checkpoint (very short)
Choose/correct the approach↓
Agent A + Agent B implement independently
↓
Automated tests/static analysis
↓
Reviewer agents analyze both solutions
↓
Reviewer produces differences + risks + evidence, not just scores
↓
Human final checkpoint
Inspect the highlighted differences/risks and test evidence → modify/accept solution
↓
Merge

This uses the human at **two high-value points**, rather than continuously supervising the agents.

The first checkpoint prevents both agents from spending time implementing the *wrong thing*. The second makes the AI reviewers **decision-support tools rather than decision-makers**. That aligns well with the successful collaboration that involves dividing work by expertise and bringing the human in for important decisions, while allowing the agent to perform routine work.

Then the **human makes the actual comparative judgment**. This preserves automation while concentrating human effort exactly where human judgment adds the most value.